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Dynamic modeling for vapor compression systems—Part II: Simulation tutorial

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Review Article

Dynamic modeling for vapor compression systems—Part II: Simulation tutorial

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This two-part article provides an introduction to dynamic modeling for vapor compression systems. Part II presents example physics-based models for each component with a discussion on common assumptions and model variations. For two-phase heat exchangers, examples for both moving boundary and finite-control volume approaches are given, along with their associated advantages and limitations. Particular modeling challenges, such as model initialization, validation, and numerical simulation, are also addressed, and sample simulation results are utilized to compare modeling paradigms and illustrate key issues. Rather than advocating the use of a particular software tool, this article provides a general tutorial on constructing dynamic simulations of vapor compression systems, outlining potential challenges and possible solutions.

Introduction

Advancements in computation, simulation, and control have served as enabling technologies for creating accurate, real-time, physics-based models of vapor compression system (VCS) dynamics. While some of the difficulties associated with VCS simulation are common to simulation efforts in general, the complexities of two-phase flows offer unique challenges and rich transient behavior. Part II of this two-part article seeks to provide a tutorial regarding dynamic simulation of VCSs using physics-based models. Issues of modeling approach, validation, integration, and computation will be addressed, as well as potential difficulties.

Defining model purpose

Modeling is an exercise in abstraction. Thus, before initiating any modeling effort, it is critical to define the desired purpose of the model so as to guide the necessary assumptions and specify the criteria for evaluating the modeling effort. Dynamic models serve a multitude of purposes, and although developing highly detailed models may seem to be the best choice to ensure accuracy, it may impede usefulness. The principle of parsimony must be employed during the process to ensure that an appropriate model is created.

Dynamic models of VCSs are valuable for (1) system analysis, design, and optimization;

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(2) model-based control design; (3) control tuning and commissioning; and (4) fault detection and diagnosis. Each of these tasks may benefit from a different modeling approach with varying levels of detail. For example, while a linearized model of low dynamic order is preferred for model-based multi-variable control design, a detailed nonlinear simulation model may be preferred for tuning a single proportional-integral-derivative (PID) controller or optimizing overall system design. Readers are encouraged to keep this perspective as they evaluate and select an appropriate modeling/simulation approach.

Overview of dynamic modeling approaches

As discussed in part I of this article, dynamic models are often classified as physics-based and data-based models. While data-based models are an effective and fast method of generating dynamics models, they are valid for only the system and scenario from which the data was obtained. Thus simulations of VCSs generally employ models based on fundamental physics to ensure applicability to a wide range of conditions and modularity in system design and configuration. For this reason, the following discussion is restricted to physics-based models; for an overview of data-based modeling techniques and appropriate references, the reader is referred to part I of this article.

Notation

The Nomenclature section located at the end of this article defines the notation of variables used throughout the article. To the extent possible, standard notation is used. However, some exceptions are necessary, as this article spans more than one field of study, which results in some inevitable conflicts in standard notation.

Physics-based dynamic models

In this section, physics-based modeling approaches for each component of a typical VCS are presented and discussed. A VCS model is constructed by appropriately integrating models for each of the individual component models that compose the cycle: condenser, expansion valve, evaporator, and compressor (Figure 1). The primary focus is on two-phase heat exchanger dynamics that de-

termine the evolution of system pressures with inlet/outlet mass flow rates as model inputs. Mass flow devices, such as compressors and valves, are also discussed briefly, as these models are significantly simpler and utilize inlet and outlet pressures to determine mass flow rates. Outlet fluid enthalpy is determined by each component model and supplied as an input to subsequent component models. Receiver models are addressed separately. After individual component models are reviewed, fundamental theoretical and practical challenges associated with VCS model simulation and prediction is discussed.

Compressors

The bulk of residential, commercial, and industrial VCSs are driven by positive displacement compressors. Except for the thermal capacitance of the compressor shell, the dynamics of these compressors typically evolve on much faster time scales than the heat exchanger dynamics, and they are thus modeled with static relationships that dictate the refrigerant mass flow rate and the outlet fluid enthalpy. This approach is typical of the majority of publications regarding VCS dynamics. The mass flow rate (Equation 1) is characterized by the speed, ω_k , the volume, V_k , the inlet density $\rho_k = \rho(P_{k,in}, h_{k,in})$ and a volumetric efficiency, typically modeled using a semi-empirical relationship based on the pressure ratio and speed $\eta_{vol} = f(P_{k,out}/P_{k,in}, \omega_k)$. The state of the outlet fluid is determined assuming an adiabatic efficiency (Equation 2), where the isentropic enthalpy is determined as

$$\begin{aligned} h_{out,isen} &= h(P_{k,out}, s_{k,in}) \text{ and} \\ s_{k,in} &= s(P_{k,in}, h_{k,in}), \\ \dot{m}_k &= \omega_k V_k \rho_k \eta_{vol}, \end{aligned} \quad (1)$$

$$\frac{h_{k,out,isen} - h_{k,in}}{h_{k,out} - h_{k,in}} = \eta_k. \quad (2)$$

As needed, a lumped capacitance model of the compressor shell thermal dynamics may be included.

Expansion valves

Expansion valves are a primary means of metering refrigerant flow in a VCS. Orifice and capillary valves have a fixed area and are designed to meter the proper flow of refrigerant at a particular condition. More complex valves utilize mechanical

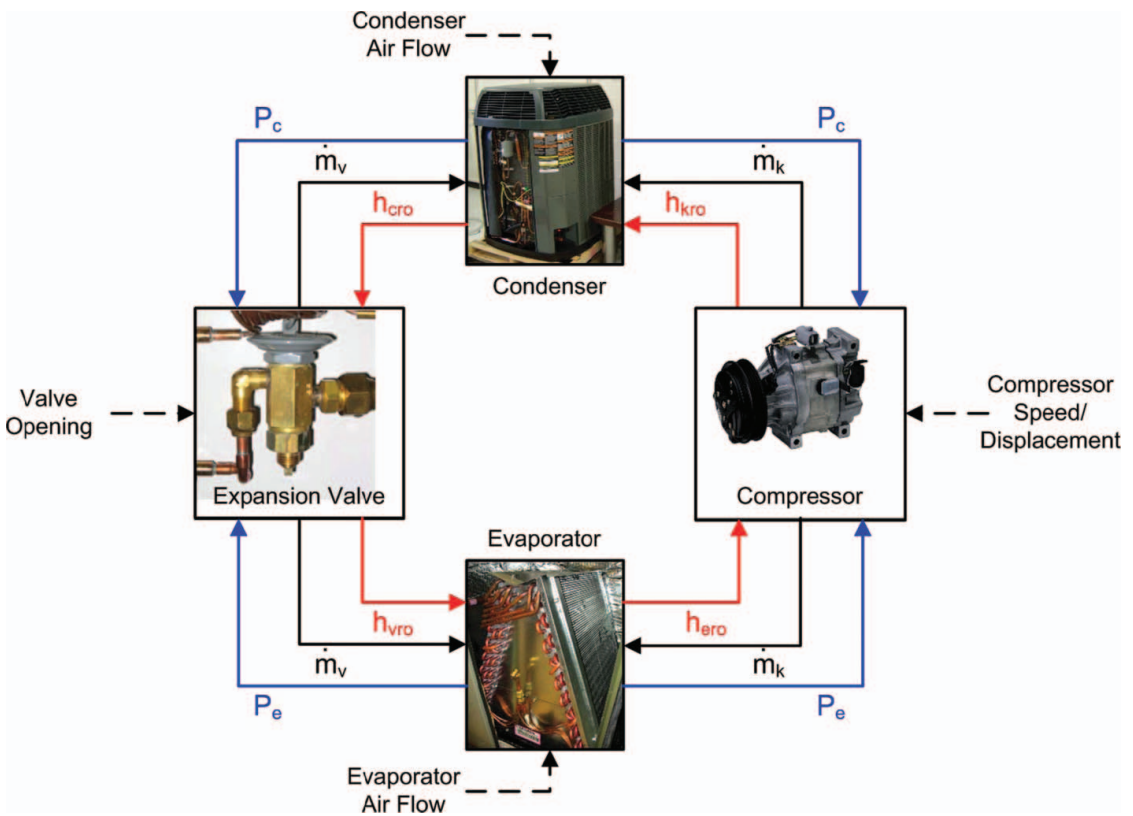


Figure 1. System modeling: external inputs and interconnection variables (color figure available online).

or electronic means to regulate refrigerant flow based on pressure or temperature. Thermostatic expansion valves (TEVs) utilize a sensing bulb filled with a two-phase refrigerant placed at the evaporator outlet to measure superheat temperature. The bulb pressure acts on a diaphragm inside the valve to increase/decrease the flow area accordingly. Expansion valves that alter the flow area to regulate pressure, rather than superheat, are termed pressure regulation valves, or automatic expansion valves (AEVs). These work on the same principle as the TEV, but instead of a sensing bulb, they use the evaporator pressure directly. Electronic expansion valves modify the flow area using an externally controlled mechanism, typically a stepper motor.

In virtually all published models, these valves are assumed to be ideal throttling valves, and thus isenthalpic, $h_{v,in} = h_{v,out}$. For fixed orifice expansion valves, the refrigerant flow rate is typically modeled using a form of Bernoulli's equation (Equation 3), where $\rho_v = \rho(P_{v,in}, h_{v,in})$. The discharge coefficient is determined empirically or from manufacturer information, $C_d = f(A_v, \Delta P)$.

Characterization of the orifice area depends on the type of valve. For fixed orifice valves, the area is constant, $A_v = a_0$; whereas for electronic expansion valves, the orifice area is directly controlled, $A_v = f(u_v)$. In TEVs, the orifice area depends on internal geometry and is adjusted by fluctuating bulb and evaporator pressures, $A_v = f(P_{bulb}, P_e)$. Pressure in the sensing bulb is typically modeled by performing an energy balance (Equation 4), leading to a simple first-order dynamic model (Equation 5) with a time constant τ . By including separate energy balance equations for the bulb, pipe wall, refrigerant, etc., it is possible to develop higher-order models of TEV behavior:

$$\dot{m}_v = A_v C_d \sqrt{\rho_v (P_{v,in} - P_{v,out})}, \quad (3)$$

$$m_{bulb} C_{p,bulb} \dot{T}_{bulb} = \alpha_o A_o (T_{amb} - T_{bulb}) - \alpha_i A_i (T_{bulb} - T_{ero}), \quad (4)$$

$$\frac{T_b(s)}{T_{ero}(s)} = \frac{k_o}{\tau s + 1}. \quad (5)$$

Receivers/accumulators

Receivers and accumulators are used in VCSs to store excess refrigerant and ensure that the fluid leaving the pressure vessel is in the appropriate fluid phase (e.g., refrigerant leaving low-side accumulator and entering compressor is a vapor). Modeling of these components is straightforward, applying conservation of mass and energy to the refrigerant and conservation of energy to the receiver shell (Estrada-Flores et al. 2003).

$$\dot{m}_{rec} = \dot{m}_{rec,in} - \dot{m}_{rec,out}, \quad (6)$$

$$\frac{d(m_{rec}u_{rec})}{dt} = \dot{m}_{rec,in}h_{rec,in} - \dot{m}_{rec,out}h_{rec,out} - \alpha_i A_i (T_{rec} - T_{rec,w}), \quad (7)$$

$$(C_p m)_{rec,w} \dot{T}_{rec,w} = \alpha_i A_i (T_{rec} - T_{rec,w}) - \alpha_o A_o (T_{rec,w} - T_{amb}). \quad (8)$$

For simulation, these equations are expanded, recognizing that the receiver volume is fixed, $V_f + V_g = V_{rec}$, by using relationships that define the refrigerant mass and energy in terms of their saturated liquid and vapor phases, $m_{rec} = \rho_f V_f + \rho_g V_g$ and $m_{rec}u_{rec} = \rho_f V_f u_f + \rho_g V_g u_g$. In this way, all the time derivatives of saturated quantities can be defined in terms of a single dynamic variable: the receiver pressure, i.e., $\dot{u}_f = (\frac{du_f}{dP})\dot{P}_{rec}$. Depending on the placement of the receiver on the high/low pressure side of the cycle, the exit enthalpy is known to be either saturated liquid or vapor, respectively.

Computational difficulties may arise when combining pressure vessels with heat exchanger models, as the intermediate mass flow rate is dictated by the small pressure differential between the two components. This tight coupling leads to stiff differential equations (see subsequent section). Thus, some researchers propose integrating the heat exchanger and receiver models, assuming isobaric conditions and neglecting the intermediate pressure drop (Eldredge et al. 2008).

Heat exchangers

As discussed in part I of this article, the thermal dynamics of a VCS are typically slower than the mechanical dynamics, and the bulk of the model complexity lies in the two-phase flow heat exchangers. The majority of recent modeling and simulation efforts are segregated into either “moving bound-

ary” (MB) modes, where parameters are lumped in regions defined by fluid phase, but the fluid phase transition point is a dynamic variable, or “finite-control volume” (FCV) models that discretize the heat exchanger into many zones or cells and apply conservation equations. Example models for each paradigm are presented here with simulation results.

For convenience, all heat exchanger models can be placed in the nonlinear descriptor form given in Equation 9. In this form, x represents the dynamic states with the associated time derivatives $\dot{x}$, and the external inputs are represented by u . The vector $f(x, u)$ represents the mass and energy flows that drive the dynamic response (e.g., Equation 10). At equilibrium, $\dot{x} = 0$, and the resulting equation $f(x, u) = 0$ represents the steady-state conservation equations for refrigerant energy, refrigerant mass, and energy stored in the heat exchanger metal. The matrix $Z(x)$ is a transformation from variables of intuitive importance (e.g., refrigerant mass storage rate) to variables that facilitate simulation and are easily measured (e.g. refrigerant pressure, temperature):

$$Z(x)\dot{x} = f(x, u), \quad (9)$$

$$f(x, u) = \begin{bmatrix} \dot{m}_{in}h_{in} - \dot{m}_{out}h_{out} - \alpha_i A_i (T_r - T_w) \\ \dot{m}_{in} - \dot{m}_{out} \\ \alpha_i A_i (T_r - T_w) - \alpha_o A_o (T_w - T_{air}) \end{bmatrix}. \quad (10)$$

MB models

The MB approach seeks to capture the salient dynamics of multiple fluid phase heat exchangers while minimizing the number of differential equations required for simulation. This approach assumes a time-varying boundary between regions of different fluid states (i.e., subcooled liquid, two-phase, or superheated vapor). Separate control volumes are considered for each of the fluid regions, and the necessary distributed parameters are lumped for each of these regions (Figure 2). To derive governing differential equations suitable for simulation or analysis, the most common method is to begin with the governing partial differential equations (PDEs) for fluid flow in a tube (Grald and MacArthur 1992). Assumptions of isobaric phase change, negligible axial refrigerant conduction, and unidirectional flow are applied, and the PDEs are integrated along the length of the heat exchanger

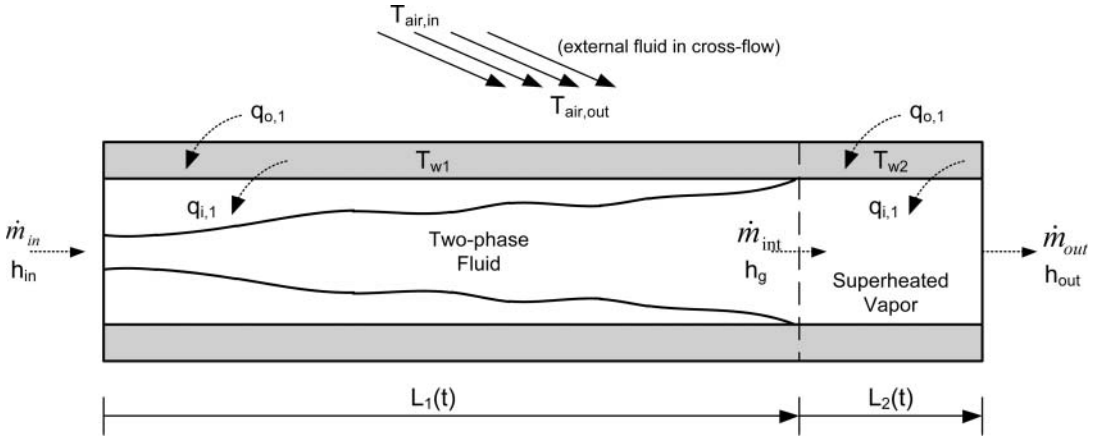


Figure 2. MB evaporator model.

to remove the spatial dependence, resulting in several ordinary differential equations (ODEs) (He et al. 1997; Rasmussen and Alleyne 2004).

As an example, for an evaporator, Equations 11 and 12 are the conservation of refrigerant mass for each region, Equations 13 and 14 are the conservation of refrigerant energy, and Equations 15 and 16 are the conservation of heat exchanger wall energy. In these equations, q is the heat transfer to/from the wall, such as $q_{i1} = \alpha_{i1} A_i \lambda_1 (T_{w1} - T_{r1})$ and $q_{o2} = \alpha_o A_o \lambda_2 (T_{air} - T_{w2})$, and λ is the normalized length of each fluid region, i.e., $\lambda_2 = L_2/L_T$:

$$\frac{d\rho_1}{dP_e} A_{cs} L_1 \dot{P}_e + (\rho_1 - \rho_g) A_{cs} \dot{L}_1 = \dot{m}_i - \dot{m}_{int}, \quad (11)$$

$$\frac{d\rho_2}{dP_e} A_{cs} L_2 \dot{P}_e + \frac{d\rho_2}{dh_o} A_{cs} L_2 \dot{h}_o + (\rho_g - \rho_2) A_{cs} \dot{L}_1 = \dot{m}_{int} - \dot{m}_o, \quad (12)$$

$$\left(\frac{d\rho_1 h_1}{dP_e} - 1 \right) A_{cs} L_1 \dot{P}_e + (\rho_1 h_1 - \rho_g h_g) A_{cs} \dot{L}_1 = \dot{m}_i h_i - \dot{m}_{int} h_g + q_{i1}, \quad (13)$$

$$\left(\frac{d\rho_2 h_2}{dP_e} - 1 \right) A_{cs} L_2 \dot{P}_e + \frac{d\rho_2 h_2}{dh_o} A_{cs} L_2 \dot{h}_o + (\rho_g h_g - \rho_2 h_2) A_{cs} \dot{L}_1 = \dot{m}_{int} h_g - \dot{m}_o h_o + q_{i2}, \quad (14)$$

$$(mC_p)_w \left(\left(\frac{T_{w1} - T_{w,int}}{L_T} \right) \dot{L}_1 + \lambda_1 \dot{T}_{w1} \right) = q_{o1} - q_{i1}, \quad (15)$$

$$(mC_p)_w \left(\left(\frac{T_{w,int} - T_{w2}}{L_T} \right) \dot{L}_1 + \lambda_1 \dot{T}_{w2} \right) = q_{o2} - q_{i2}. \quad (16)$$

The final form of governing equations differs slightly in the literature, depending on assumptions regarding the lumped fluid properties, void fraction, and wall temperature at the interface between the regions, $T_{w,int}$. For example, the properties for the two-phase region typically employ a mean void fraction $\rho_1 h_1 = \rho_f h_f (1 - \bar{y}) + \rho_g h_g (\bar{y})$, the properties in the single-phase region are generally lumped using an average value $h_2 = \frac{1}{2}(h_g + h_o)$, and the interface wall temperature is assumed to simply be equal to the two-phase wall temperature $T_{w,int} = T_{w1}$. These six equations can be algebraically combined to eliminate the mass flow rate at the fluid interface and then organized into the $Z(x)\dot{x} = f(x, u)$ form, as in Equation 17. The matrix $Z(x)$ remains full rank and invertible as long as $\lambda_{1,2} > 0$, or in other words, as long as evaporator superheat is maintained:

$$\begin{bmatrix} z_{11} & z_{12} & 0 & 0 & 0 \\ z_{21} & z_{22} & z_{23} & 0 & 0 \\ z_{31} & z_{32} & z_{33} & 0 & 0 \\ 0 & 0 & 0 & z_{44} & 0 \\ z_{51} & 0 & 0 & 0 & z_{55} \end{bmatrix} \begin{bmatrix} \dot{L}_1 \\ \dot{P}_e \\ \dot{h}_o \\ \dot{T}_{w1} \\ \dot{T}_{w2} \end{bmatrix} = \begin{bmatrix} \dot{m}_i (h_i - h_g) + q_{i1} \\ \dot{m}_o (h_g - h_o) + q_{i2} \\ \dot{m}_i - \dot{m}_o \\ q_{o1} - q_{i1} \\ q_{o2} - q_{i2} \end{bmatrix}. \quad (17)$$

While this is a distinctly nonlinear model, constructing a linear approximation for the purposes of dynamic analysis and control design is straightforward. First the values of the state variables are determined at the desired steady-state operating point, $x_o = [L_1(0) \ P_e(0) \ h_o(0) \ T_{w1}(0) \ T_{w2}(0)]$, as are the values of the external inputs, $u_o = [\dot{m}_i(0) \ \dot{m}_o(0) \ h_i(0) \ \alpha_o(0) \ T_{air}(0)]$. The linearized model is then formulated in terms of the deviation variables $\delta x = x - x_o$, and $\delta u = u - u_o$ as:

$$\delta \dot{x} = \left[Z(x_o)^{-1} \frac{\partial f(x_o, u_o)}{\partial x} \right] \delta x + \left[Z(x_o)^{-1} \frac{\partial f(x_o, u_o)}{\partial u} \right] \delta u. \quad (18)$$

Additional details regarding this linearization process, and a complete description of the resulting model, can be found in several places in the literature (He et al. 1998; Rasmussen and Alleyne 2004). The linearized model is a local approximation, and the approximation errors grow as the model operates away from the operating point of linearization. Although nonlinearities in the heat exchanger parameters (e.g., heat transfer correlations) and fluid properties do result in some error, the dynamics remain qualitative the same (Figure 3).

Fixed-control volumes models

Finite-difference or FCV approaches offer the capability to model the fluid behavior with extreme detail, capturing the thermo-physical gradients and distributed parameters. This, reportedly, lends itself to higher accuracy prediction than the

lumped parameter approaches that rely on effective parameters and only describe the dominant physical mechanisms for heat transfer, fluid flow, etc. However, the resulting model contains many differential equations, which increases computation time, with the possibility of additional numerical challenges (e.g., stiff dynamics, differential algebraic equations [DAEs], iterative solvers, etc.). If a pressure gradient is assumed, then a staggered grid approach for modeling the mass and energy flow for each control region is typical (Eborn 2001), where pressure differentials dictate mass flow rates, and mass flow rate differentials drive the pressure and energy dynamics (Figure 4).

The governing equations for each region include (1) the conservation of refrigerant mass (Equation 19), where the intermediate mass flow rates are determined using a manipulated form of the Darcy-Weisbach equation (Equation 20), where the coefficient C_d is a function of the tube diameter, length, friction factor, etc.; (2) the conservation of refrigerant energy (Equation 21), where the storage rate of specific energy $\dot{u}_{e,j}$ is expanded in terms of pressure and enthalpy; and (3) the conservation of heat exchanger wall energy (Equation 22), where $q_{i,j} = \alpha_{i,j} A_{i,j} (T_{w,j} - T_{r,j})$ and $q_{o,j} = \alpha_{o,j} A_{o,j} (T_{air,j} - T_{w,j})$.

$$\dot{m}_{e,j} = \dot{m}_{j-1} - \dot{m}_j, \quad (19)$$

$$\dot{m}_j = C_d A \sqrt{\rho(P_j - P_{j+1})}, \quad (20)$$

$$\begin{aligned} \dot{U}_j &= \dot{m}_{e,j} u_{e,j} + m_{e,j} \left(\frac{du_{e,j}}{dP_e} \dot{P}_j + \frac{du_{e,j}}{dh_j} \dot{h}_j \right) \\ &= \dot{m}_{j-1} h_{j-1} - \dot{m}_j h_j + q_{i,j}, \end{aligned} \quad (21)$$

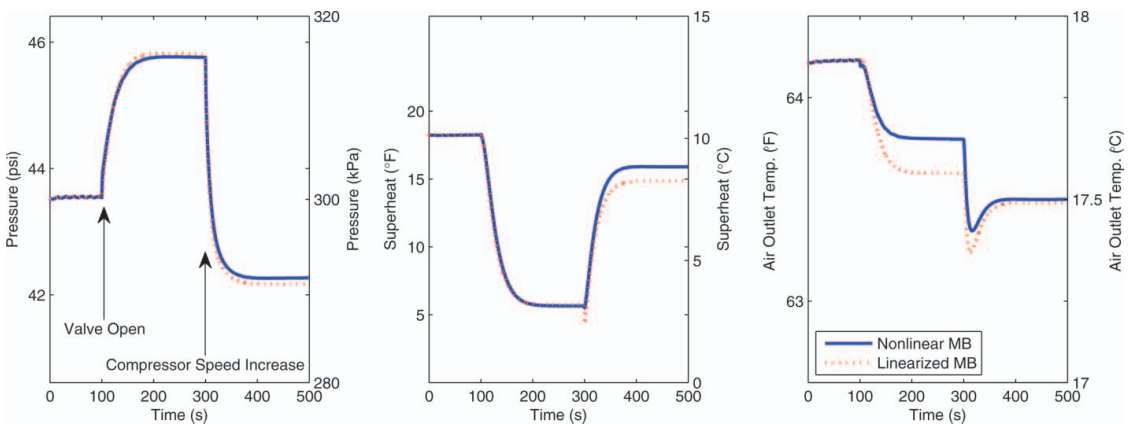


Figure 3. Comparison of nonlinear and linearized MB models for step changes in expansion valve and compressor speed. Signals shown are evaporator pressure, evaporator superheat, and evaporator air outlet temperature (color figure available online).

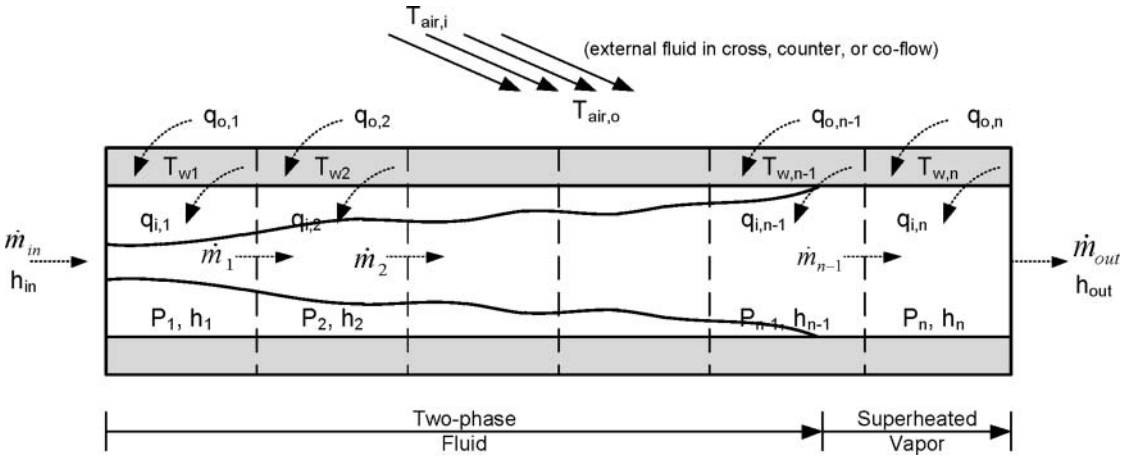


Figure 4. FCV evaporator model.

$$\dot{E}_{w,j} = (mC_p)_w \dot{T}_{w,j} = q_{o,j} - q_{i,j}. \quad (22)$$

When organized into the standard $Z(x)\dot{x} = f(x, u)$ form,

$$\underbrace{\begin{bmatrix} \dot{U}_1 \\ \vdots \\ \dot{U}_n \\ \dot{m}_{e,1} \\ \vdots \\ \dot{m}_{e,n} \\ \dot{E}_{w,1} \\ \vdots \\ \dot{E}_{w,n} \end{bmatrix}}_{\text{mass and energy storage rates}} = \underbrace{\begin{bmatrix} Z_{11} & Z_{12} & 0 \\ Z_{21} & Z_{22} & 0 \\ 0 & 0 & Z_{33} \end{bmatrix}}_{\text{pseudo-state transformation matrix}} \underbrace{\begin{bmatrix} \dot{P}_1 \\ \vdots \\ \dot{P}_n \\ \dot{h}_1 \\ \vdots \\ \dot{h}_n \\ \dot{T}_{w,1} \\ \vdots \\ \dot{T}_{w,n} \end{bmatrix}}_{\text{dynamic state variables}} \quad (23)$$

$$= \underbrace{\begin{bmatrix} \dot{m}_i h_i - \dot{m}_1 h_1 + q_{i,1} \\ \vdots \\ \dot{m}_{n-1} h_{n-1} - \dot{m}_o h_o + q_{i,n} \\ \dot{m}_i - \dot{m}_1 \\ \vdots \\ \dot{m}_{n-1} - \dot{m}_o \\ q_{o,1} - q_{i,1} \\ \vdots \\ q_{o,n} - q_{i,n} \end{bmatrix}}_{\text{steady state mass and energy balances}} \quad (23)$$

If isobaric conditions are assumed for the heat exchanger, $P_j = P_e, \forall j$, and the conservation of momentum is neglected, the conservation equations may be simplified to eliminate the intermediate mass flow variables as $\dot{m}_j = \dot{m}_{in} - \sum_{k=1}^j \dot{m}_{e,k}$. The conservation of refrigerant energy (Equation 21) now becomes

$$\begin{aligned} \dot{U}_j &= \dot{m}_{e,j} u_{e,j} + m_{e,j} \left(\frac{du_{e,j}}{dP_e} \dot{P}_e + \frac{du_{e,j}}{dh_j} \dot{h}_j \right) \\ &= \left(\dot{m}_{in} - \sum_{k=1}^{j-1} \dot{m}_{e,k} \right) (h_{j-1} - h_j) \\ &\quad + \dot{m}_{e,j} h_j + q_{i,j}, \end{aligned} \quad (24)$$

where the rate of mass storage in each region is also expanded as $\dot{m}_{e,j} = \dot{\rho}_{e,j} V_j = \left(\frac{d\rho_{e,j}}{dP_e} \dot{P}_e + \frac{d\rho_{e,j}}{dh_j} \dot{h}_j \right) V_j$. After these substitutions, the conservation equations in Equation 25 are transformed into Equation 26:

$$\underbrace{\begin{bmatrix} \dot{U}_1 \\ \vdots \\ \dot{U}_n \\ \dot{m}_e \\ \dot{E}_{w,1} \\ \vdots \\ \dot{E}_{w,n} \end{bmatrix}}_{\text{mass and energy storage rates}} = \underbrace{\begin{bmatrix} \dot{m}_i h_i - \dot{m}_1 h_1 + q_{i,1} \\ \vdots \\ \dot{m}_{n-1} h_{n-1} - \dot{m}_o h_o + q_{i,n} \\ \dot{m}_{in} - \dot{m}_{out} \\ q_{o,1} - q_{i,1} \\ \vdots \\ q_{o,n} - q_{i,n} \end{bmatrix}}_{\text{steady state mass and energy balances}}, \quad (25)$$

$$\underbrace{\begin{bmatrix} Z_{11} & Z_{12} & 0 \\ Z_{21} & Z_{22} & 0 \\ 0 & 0 & Z_{33} \end{bmatrix}}_{\text{pseudo-state transformation matrix}} \underbrace{\begin{bmatrix} \dot{P}_e \\ \dot{h}_1 \\ \vdots \\ \dot{h}_n \\ \dot{T}_{w,1} \\ \vdots \\ \dot{T}_{w,n} \end{bmatrix}}_{\text{dynamic state variables}} = \underbrace{\begin{bmatrix} \dot{m}_{in}(h_{in} - h_1) + q_{i,1} \\ \vdots \\ \dot{m}_{in}(h_{n-1} - h_{out}) + q_{i,n} \\ \dot{m}_{in} - \dot{m}_{out} \\ q_{o,1} - q_{i,1} \\ \vdots \\ q_{o,n} - q_{i,n} \end{bmatrix}}_{\text{modified steady state mass and energy balances}} \quad (26)$$

This particular model has $2n + 1$ dynamic states, where n is the number of control volumes or cells (conservation of refrigerant/heat exchanger energy for each region, and one equation for the conservation of refrigerant mass). If conservation of momentum is included, the model includes $3n$ dynamic equations, with the additional algebraic constraints that define the mass flow rate between regions. Publications typically report using $n > 20$ regions to achieve grid (or mesh) independence. Figure 5 presents simulation results for FCV models with increasing numbers of regions. While pressure and

air temperature predictions converge for a modest level of discretization, the prediction of refrigerant outlet temperature (superheat) clearly requires $n > 20$ before the simulations show quantitative convergence.

While $n > 20$ may be required for numerical consistency, the dominant dynamics appear to be of much lower order. This indicates some level of “over-modeling,” as data-driven models do not require 50+ dynamic states to reproduce the required transient response (Rasmussen et al. 2005). In this form, these models capture subtle parameter variations but are of high dynamic order, making them most useful for control evaluation or tuning, rather than dynamic analysis or model-based control design. If such design tasks are attempted, numerical model reduction techniques are generally applied to obtain a simple low-order approximate model. Although computation time was reported to be a serious concern in some earlier publications (e.g., Wang and Toubert 1991), computer and software advances are rapidly eliminating this as a significant issue. Current capabilities allow real-time (or better) simulation of VCS models with multiple FCV heat exchangers.

When the FCV and MB models are initialized with the same physical parameters, the resulting simulations exhibit only small differences (Figure 6). The MB model responds slightly faster than the FCV model to step changes in expansion valve or compressor, indicating that some adjustment of the “effective” parameters used by the MB model may be appropriate (e.g., the user may opt to use a weighted average of a particular parameter versus strict average when lumping the parameters for

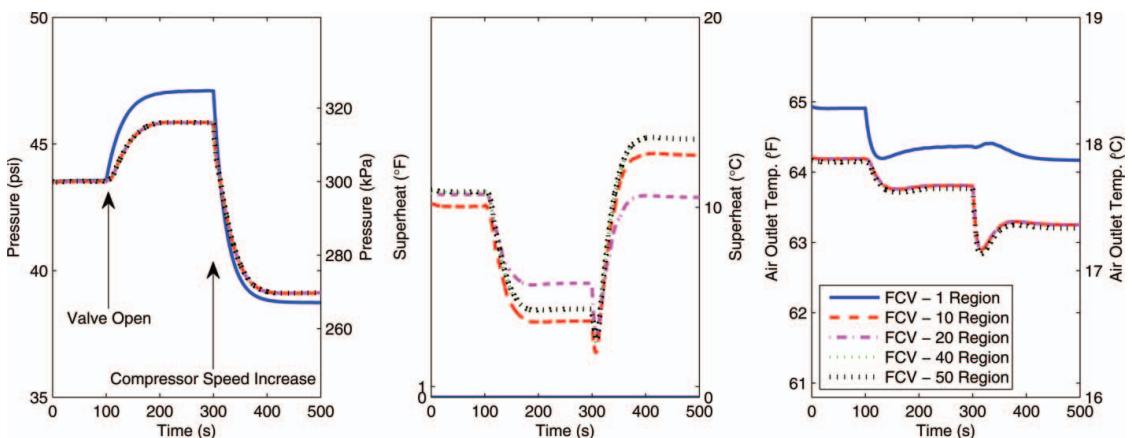


Figure 5. Comparison of FCV with differing levels of discretization for step changes in expansion valve and compressor speed. Signals shown are evaporator pressure, evaporator superheat, and evaporator air outlet temperature (color figure available online).

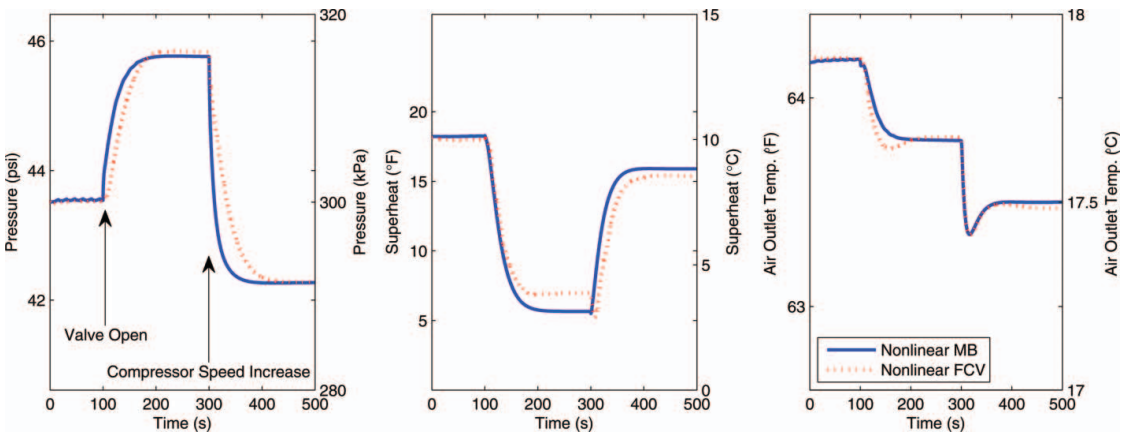


Figure 6. Comparison of FCV and MB models for step changes in expansion valve and compressor speed. Signals shown are evaporator pressure, evaporator superheat, and evaporator air outlet temperature (color figure available online).

two-phase or superheated fluid regions). The models also give slightly different steady-state predictions for evaporator outlet conditions (superheat). This discrepancy demonstrates the fundamental difference between the modeling paradigms, as the FCV model avoids extensive lumping of parameters and captures parameter and temperature gradients that result in a more precise prediction of outlet conditions. However, it is noted that these prediction errors are typically within the differences seen when validating the models against experimental data and, therefore, may not be a critical factor in selecting which model to use.

Modeling challenges and innovations

Stiff dynamics and DAEs

VCSs exhibit dynamics that evolve on several time scales. At the extremes are the dynamics of the building or cabin cooling load whose dynamic response is on the order of hours or days, whereas the propagation of pressure waves through the refrigerant is completed in milliseconds. The user typically isolates a time scale of primary interest and approximates slower dynamics as constant, and faster dynamics as instantaneous, resulting in a set of DAEs. However, isolating dynamics at particular time scales is not always possible, and the resulting system retains the multi-time scale behavior and is referred to as a “stiff” set of ODEs. Most simulation software packages include variable time-step numerical integration algorithms for stiff systems and algorithms for solving simple DAEs (i.e., DAEs of index 1). However, compiling simulations for real-time or embedded computing requires fixed step

solvers, thus requiring the user to resolve the issue of numerical stiffness prior to simulation.

The primary source of multi-time scale behavior is the inclusion of the conservation of momentum in the dynamic equations. Given that the pressure in the system propagates much faster than the evolution of the thermal dynamics, this results in a natural conflict. A common approach of addressing this issue is to assume isobaric conditions in the heat exchangers (thus neglecting the associated momentum change). Zhang et al. (2009) presented a direct comparison of the most common assumptions regarding how the momentum equation is handled, revealing that including momentum effects is not critical for large transient simulations but that discernible differences are present for instantaneous step changes in mass flow rate. However, as most compressors and valves do not respond instantly, this may not be a critical effect to capture.

Beyond momentum dynamics, most two-phase heat exchangers exhibit multi-time scale behavior. An analysis of the eigenvalues of the linearized heat exchanger dynamics reveals dynamic modes whose time scale varies by several orders of magnitude (Rasmussen and Alleyne 2004). This leads naturally to the search for a reduced-order model that eliminates dynamics outside of the time scale of interest, serving both to simplify the model computationally for simulation tasks and analytically for control design tasks. The particular method of model reduction and the physical interpretation varies in the literature. He et al. (1997) used both pure numerical reduction as well as a physically motivated technique for removing non-essential modes (He et al.

1998), an approach later duplicated by Leducq et al. (2003). Rasmussen and Alleyne (2004) explored alternative choices of state variables to identify those that virtually decoupled the modes according to time scale and then applied singular perturbation techniques for model reduction. Regardless of the approach, these studies indicate that some commonly modeled dynamics are not essential for accurate transient simulation.

Modelers of centrifugal compressors should also be aware of the potential challenges specific to the “surge-and-stall” dynamics inherent to this particular component. Models of this phenomena are readily available in the literature (see Moore and Greitzer, 1986) and numerous subsequent publications). These unstable dynamics pose a particular challenge for simulation but also typically contribute to the “stiff systems” problem, as these dynamics are much faster than heat exchanger dynamics. Integrating standard VCS models with centrifugal compressors is likely to be an area of research in the near future, specifically for aircraft cooling systems.

Real-time simulation

Computation speed for VCS simulations is largely dependent on the chosen heat exchanger modeling paradigm. As described above, solving the nonlinear differential equation $Z(x)\dot{x} = f(x, u)$ requires a matrix inversion, where the matrix size depends on the number of dynamic states. MB models typically have five to nine states, depending on the formulation, and FCV models have $2n + 1$ dynamic states, where n is the number of control volumes. Linearized model approximations are capable of much faster simulation, as the matrix inversion is only computed once during the initialization of the simulation to compute the appropriate state space matrices (Equation 18). Compiled code also offers a significant improvement in execution time, although this eliminates the option of modifying the simulation or viewing the predicted outputs during execution.

For comparison, Table 1 summarizes the computation time required to produce the 500-second simulations shown in Figures 3 and 5. The simulations were conducted on a standard desktop computer (3-GHz processor) using a fourth-order Runge-Kutta solver with a fixed step size of 0.01 s. The “standard simulation” results were created by simulating the systems using MATLAB/Simulink with the default simulation settings (i.e., no “accelerated” simula-

tion options were used). The same simulations were also compiled into stand-alone executable files resulting in order-of-magnitude improvements in execution speed. The real-time factor was computed by dividing the total simulation time (initialization + execution) by the length of the simulated time (500 s).

Fluid properties and heat transfer correlations

Refrigerant properties pose a particular challenge for dynamic VCS simulation, as standard methods using equations of state (EOS) for determining fluid properties can be computationally expensive and not suitable for real-time dynamic simulations that require updating of fluid properties at each simulation time step. Likewise, using software interfaces to external databases or companion software programs can significantly increase simulation computation time. A typical solution to this dilemma is to utilize high-accuracy fluid property database software, e.g., NIST’s REFPROP (Lemmon et al. 2007), to generate tables of fluid properties for a predetermined range of conditions. This table is then used for fast interpolation of fluid properties during simulation. Likewise, tables for the partial derivatives of fluid properties may be generated by computing the numerical gradient or from fundamental thermodynamic relationships (e.g., Appendix B of Tummescheit 2002). Care must be taken to appropriately handle discontinuities in the property tables, such as the transition point between fluid phases, or significant numerical inaccuracies will occur during interpolation. While this method of fluid property calculation sacrifices some level of accuracy, increasing the grid density of the interpolation tables can reduce the numerical errors to insignificant levels.

Correlations for heat transfer, void fraction, and fluid flow are also the source of potential difficulties. For maximum accuracy, the values for heat transfer coefficients, pressure drop, etc. must be updated during the simulation. Some approximations may be necessary to enable fast explicit calculation of values and to avoid circular algebraic dependencies that would require calling a numerical equation solver at each time step. For example, a heat transfer coefficient correlation may require heat flux as a parameter. However, the instantaneous heat flux is determined using the heat transfer coefficient. With only minor loss in accuracy, the heat flux from the previous time step may be used as an approximation of the current heat flux for calculating the heat

Table 1. Comparison of simulation execution time.

Model type	Dynamic states	Standard simulation			Compiled simulation		
		Initialization (s)	Execution (s)	Real-time factor	Compilation (s)	Execution (s)	Real-time factor
Linearized MB	5	4.1	5.4	0.02	16.8	< 0.1	0.03
Nonlinear MB	5	10.2	21.2	0.06	28.7	0.6	0.06
Nonlinear FCV—1 Region	3	8.3	22.8	0.06	30.5	1.7	0.06
Nonlinear FCV—10 Region	21	12.8	158.7	0.34	31.9	11.1	0.09
Nonlinear FCV—20 Region	41	13.9	774.5	1.58	32.6	23.8	0.11
Nonlinear FCV—30 Region	61	14.4	2166.1	4.36	33.8	37.1	0.14
Nonlinear FCV—40 Region	81	13.2	4772.5	9.57	41.5	53.2	0.19
Nonlinear FCV—50 Region	101	14.2	9153.4	18.34	43.0	72.4	0.23
Nonlinear FCV—100 Region	201	19.0	70,594	141.23	47.5	275.2	0.65

transfer coefficient and avoiding an algebraic dependency. Although this is an obvious approximation, numerical inaccuracies are typically well within the uncertainty levels of the correlation. More important is to ensure continuity in the computation to avoid false transient responses due to discontinuous changes in model parameters during a simulation. For example, numerical problems in solving for heat transfer correlation may lead to a sudden change in the value of the coefficient. This, in turn, would induce a transient response not unlike a sudden blockage or fouling of the heat exchanger. Thus, care must be taken to ensure continuous calculation of the correlation from one simulation time step to the next.

Model validation and tuning

Central to any model-based research effort is establishing the validity of the model. The problem of model validation is separate and distinct from the issues associated with model *purposiveness* (e.g., whether the model meets the required purpose), model *plausibility* (e.g., whether the model conceptually agree with the process physics), and model *verification* (e.g., whether the software embodiment of the model is accurate) (Bohlin 1991). Model *validation* is a question of assessing whether the model outputs agree with experimental data (Murray-Smith 1998). From a philosophical perspective, it is impossible to truly validate a model, as this would require an infinite amount of experimental data (Smith et al. 1997); model *invalidation* is a more proper term, as the majority of techniques are used to determine when a model is not capable of reproducing the measurements (i.e., model falseness).

The techniques for external model validation can be considered qualitative (e.g., visual inspection of model/data agreement) and quantitative (e.g., numerical metric of model validity). Common quantitative methods for dynamic models include (1) statistical, (2) residual-based, (3) parametric, and (4) robustness methods. Statistical methods can be either descriptive statistics that evaluate the mean, variances, and correlations of the data or inferential statistics that deal with hypothesis testing (Ljung 1999). Residual-based approaches rely on numerical performance measures by calculating the root mean square of the prediction error, relative prediction error, or other variations (Murray-Smith 1998) and generally assume that model residuals are caused by noise that is uncorrelated to the input

(Ljung 1999). Parametric-based approaches explore the sensitivity of the model predictions to parameter variations and techniques for improving model fit by parameter tuning (Butterfield and Thomas 1986a, 1986; Butterfield 1990). Robust model validation is specifically designed for control-oriented models and seeks to determine the level of dynamic model uncertainty required to reproduce the experimental data (Poolla et al. 1994; Dullerud and Smith 1996; Smith and Dullerud 1996).

As seen from the discussion in previous sections, despite the large number of publications regarding VCS dynamic models, relatively few provide detailed experimental validation. The most common deficiencies are (1) no validation, (2) only steady-state validation of a dynamic model, (3) comparison with other simulation models, (4) comparison of only a few model outputs or not the critical outputs, or (5) qualitative comparison with no quantitative metrics. Depending on the intended model use, validation of the model for small transient deviations may or may not be appropriate; this includes frequency domain validation, which focuses on linear model behavior.

Perhaps the lack of detailed model validation in the literature stems from the challenge of proper initializing and tuning of the model. Most VCS models contain measurable physical parameters (areas, volumes) as well as unmeasurable or uncertain parameters (void fraction, heat transfer correlations). Additionally, many model parameters are lumped (effective surface areas, effective volumes, etc.) and must be adjusted given model assumptions. For example, whereas the effective internal surface area of a heat exchanger includes only the area that directly participates in heat transfer, the effective heat exchanger volume should be adjusted to include headers and entrance exit pipe lengths because all of these participate in the increase/decrease in system pressure. The following figures illustrate some of the key tuning parameters and their relative effects on predicted temperatures and pressures.

Figure 7 presents MB model simulations where the external heat transfer coefficient has been increased. While the pressure and superheat response remain relatively unchanged, the outlet air temperature is lowered by several degrees and exhibits a transient response that is more responsive to valve and compressor changes. Figure 8 shows the effect of increasing the effective cross-sectional area of the heat exchanger or, in other words, increasing the total internal volume. Assuming a constant void

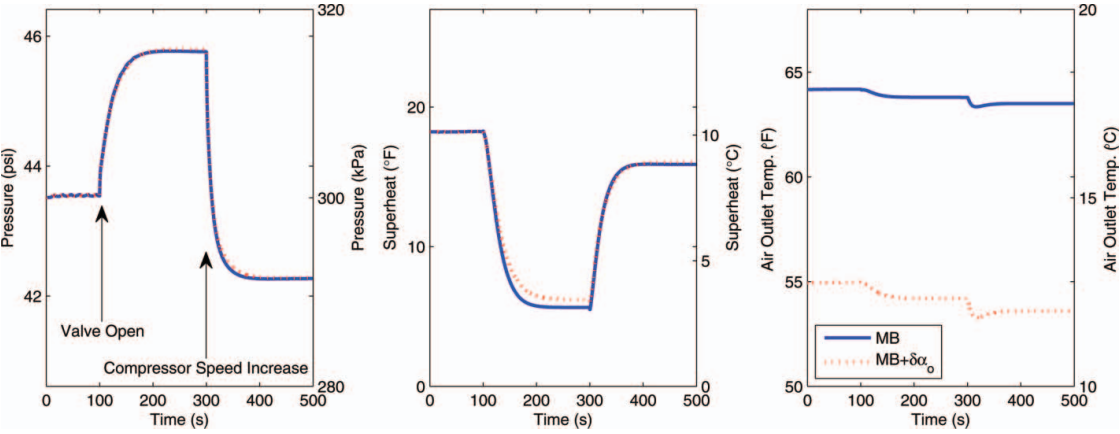


Figure 7. Comparison of MB simulations for variations in external heat transfer coefficient. Transients are induced by step changes in expansion valve and compressor speed (color figure available online).

fraction, this also equates to more refrigerant mass in the heat exchanger. As expected given the large volume to pressurize and the greater mass to affect, the transient responses are somewhat slower, and the superheat response is noticeably reduced in magnitude. Figure 9 presents an example of a parameter that affects primarily the transient response rather than the steady-state values. For void fraction correlations, the slip ratio defines the velocity of the vapor phase to liquid phase (homogeneous flow refers to a slip ratio of one). Increasing the slip ratio decreases the mean void fraction and increases the amount of refrigerant mass. Thus, for higher slip ratios, the transient response of all signals is slower.

A final comment on model tuning: VCS models exhibit a significant sensitivity to refrigerant mass flow rate prediction. Because the heat exchanger model dynamics are driven by mass flow rate differentials (inlet versus outlet), accurate valve and compressor models are critical. This is somewhat of a paradox, as the bulk of the dynamic complexities lie with the heat exchanger models, but it is the quantitative accuracy of the mass flow models that largely determines the quality of the validation and comprise much of the parameter tuning effort. Given the large number of parameters to be tuned, it is therefore essential to cross-validate the models, using different datasets to tune and validate the model.

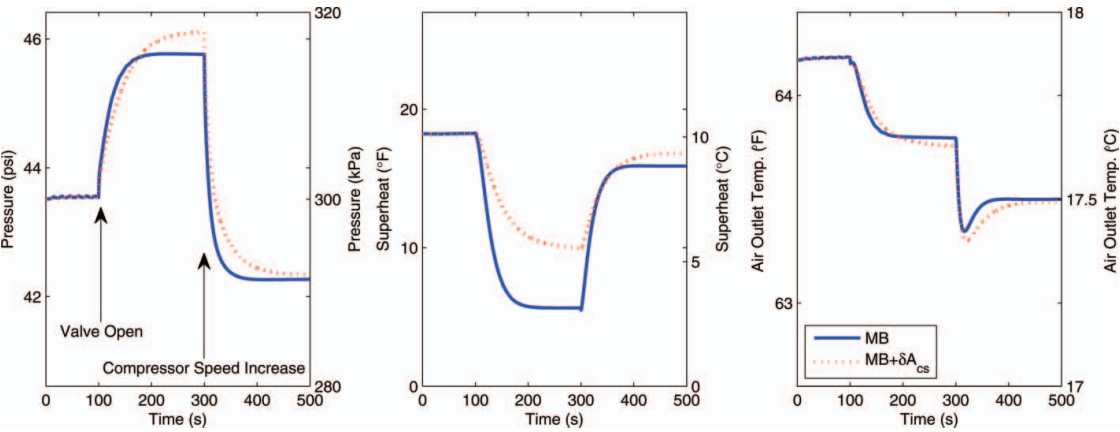


Figure 8. Comparison of MB simulations for variations in heat exchanger cross-sectional area (i.e., heat exchanger internal volume). Transients are induced by step changes in expansion valve and compressor speed (color figure available online).

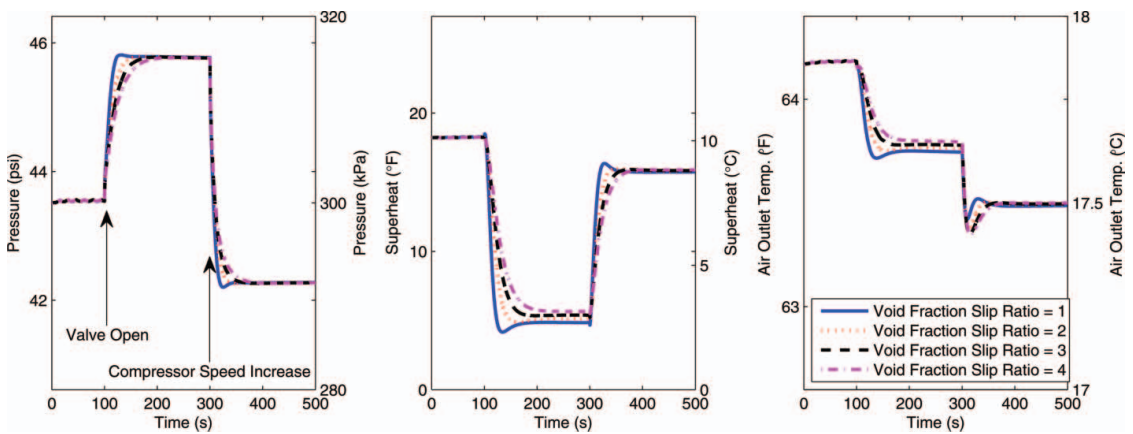


Figure 9. Comparison of MB simulations for variations in void fraction slip ratio. Transients are induced by step changes in expansion valve and compressor speed (color figure available online).

Simulation of hard transients

VCS startup and shutdown scenarios are among those termed “hard” or “large” transients in the literature. The simulation challenge posed by these scenarios is the violation of modeling assumptions and the resulting numerical singularities. For example, for the MB modeling paradigm, the loss of the evaporator superheat region results in a singularity of the $Z(x)$ matrix, which prevents inversion and the calculation of time derivatives. Before this happens, however, the model outputs become infinitely large, as the equations have terms in the denominators that approach zero (Equations 16 and 17). Overcoming this challenge led to the development of switched MB (SMB) models that handle the disappearance or introduction of a fluid phase region. Similar challenges are encountered when attempting to simulate the dynamic effect of some system faults, such as fan blockage, valve failure, or refrigerant leak. Of secondary concern is the heightened role of phenomenon typically ignored during modeling for pseudo steady-state operation (e.g., pool boiling or free convection heat transfer when fans are off, back flow or leakage when the compressor is off, and gravity-driven refrigerant pooling during system shutdown).

By assuming appropriate initial conditions, including a distribution of refrigerant that meets model assumptions, the startup scenario can be handled using standard FCV and MB models (e.g., Bendapudi and Braun 2002). However, cycling (startup and shutdown) of a VCS requires models capable of handling the hard transient effects. While earlier

efforts examine cycling from a purely experimental perspective, recent innovations in both SMB and FCV models have resulted in experimentally validated models for startup and shutdown transients (Bendapudi et al. 2008; Hermes and Melo 2008; Li and Alleyne 2010). Typically, these studies focus on matching the gross pressure or cooling capacity trends, rather than more specific transients responses like superheat. They also tend to focus on short-term shutdown, as longer off periods tend to accentuate the effect of unmodeled phenomena, such as refrigerant pooling and the resulting slug flow.

While the FCV models are intended to handle hard transients, the MB models require specific additions to handle the loss of superheat or subcooling. The published models differ in terms of the switching criteria and the assumptions regarding the heat exchanger wall temperature at the interface between fluid phases. Pettit et al. (1998) used a simple switching strategy based on comparing refrigerant exit enthalpies to saturation enthalpies to handle all cases of switching on/off superheat or subcool, and they added small numerical terms to avoid singularities when $L = 0$. Zhang and Zhang (2006) and McKinley and Alleyne (2008) handled the disappearance of superheat or subcool regions based on the length the associated region, the former simply requiring $L_{SH} > 0$ for persistence of the superheat region, and the latter using $L_{SH} < L_{min}$ and $dL_{SH}/dt < 0$ as the conditions for switching the superheat region off. Zhang and Zhang (2006) did not specify criteria for handling the cases where superheat

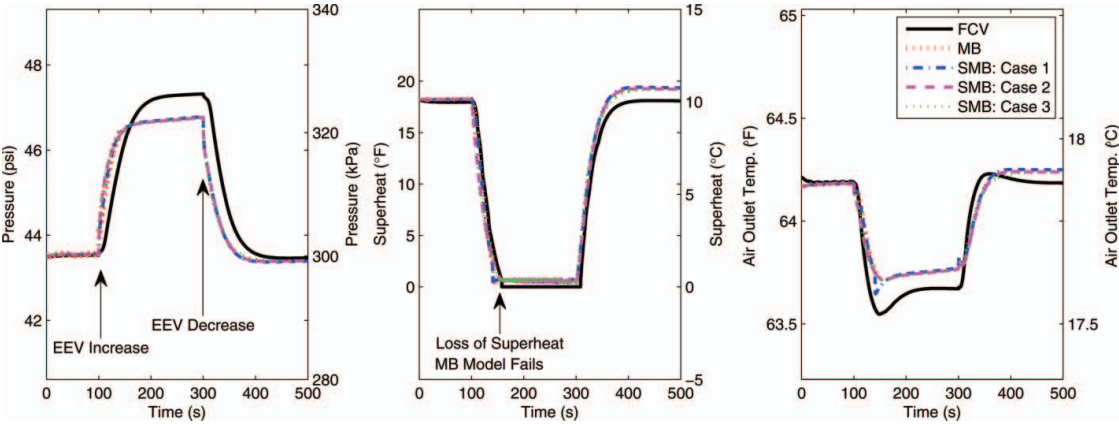


Figure 10. Comparison of FCV, MB, and SMB models for the loss of evaporator superheat induced by step changes in expansion valve. Signals shown are evaporator pressure, evaporator superheat, and evaporator air outlet temperature (color figure available online).

reappears, but McKinley and Alleyne (2008) used a criteria based on void fraction. The assumptions regarding interface wall temperatures also vary. Petit et al. (1998) made the standard assumption that the wall temperature at the interface is equal to the two-phase wall temperature (Case 1), whereas McKinley and Alleyne (2008) used an approach suggested in Jensen and Tummescheit (2002) that the interface temperature depends on whether the two-phase region is growing or shrinking, $T_{w,int} = T_1$ if $\frac{dL_1}{dt} < 0$ and $T_{w,int} = T_2$ if $\frac{dL_1}{dt} > 0$ (Case 2). Zhang and Zhang (2006) assumed a weighted average $T_{w,int} = \frac{L_2}{L_{total}} T_1 + \frac{L_1}{L_{total}} T_2$, which eliminated the numerical singularity associated with dividing by the length of a disappearing region (Case 3).

The following figures illustrate the behavior of these various models under hard transients. The simulation shown in Figure 10 assumes that the expansion valve is opened wide for a period of time, resulting in a large increase in refrigerant flow into the evaporator. Without changing the airflow rate or the compressor speed, this results in a loss of evaporator superheat. The MB model fails, with the prediction values exploding toward infinite due to the singularity in the $Z(x)$ matrix that makes it not invertable. However the FCV and SMB models both handle the transient without numerical failures. Although examination of the internal model variables, such as wall temperature, differ distinctly depending on the assumptions regarding

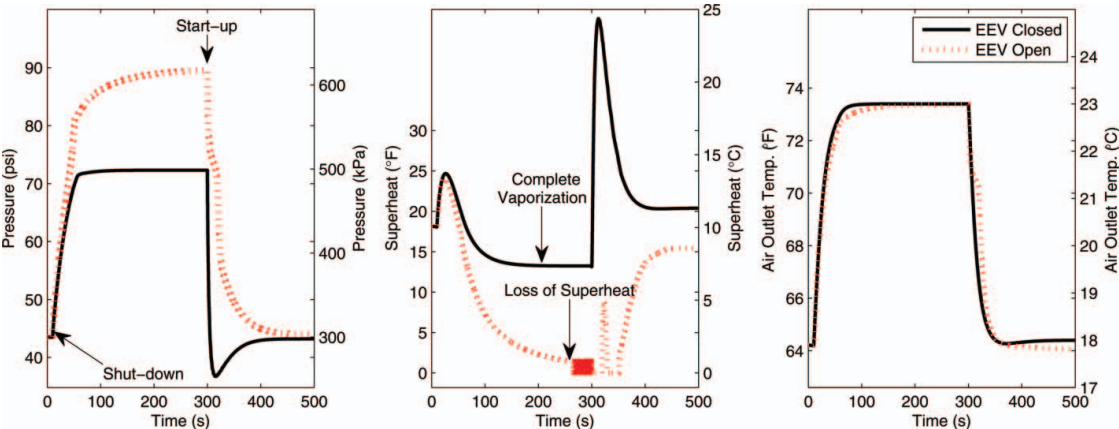


Figure 11. Comparison of expansion valve strategies during compressor shutdown using FCV models. Signals shown are evaporator pressure, evaporator superheat, and evaporator air outlet temperature (color figure available online).

the interface wall temperature, the principal output signals are virtually indistinguishable.

Figure 11 presents simulation results for startup and shutdown transients for a simple air-cooling evaporator with an electronic expansion valve and a reciprocating compressor. The simulation considers two scenarios: (1) the expansion valve closes simultaneously with the compressor and (2) the expansion valve is open during compressor shutoff. Note that for the former case, complete vaporization of the refrigerant in the evaporator is achieved. Additionally, note that for the latter case there is a slight discontinuity, as the last region transitions from two phases to single phase. This discontinuity is decreased with finer discretization of the heat exchanger.

Conclusion

Recent developments in techniques and software significantly expand the range of tools for dynamic modeling of VCSs. Users should carefully select the appropriate modeling paradigm based on the intended use of the model. Because of the uncertain nature of some VCS model parameters, experimental validation is integral to every modeling effort. While existing models can capture salient VCS dynamics in the majority of scenarios, continuing efforts in the areas of model reduction, numerical solvers, and automated model tuning and validation are necessary to increase the capabilities of these models for analysis, design, control design, and fault detection.

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Nomenclature

Thermodynamic variables

E	= energy
P	= pressure
T	= temperature
U	= internal energy
h	= specific enthalpy
$\dot{m}$	= mass flow rate

q	= heat
s	= specific entropy
u	= specific internal energy
α	= heat transfer coefficient
$\bar{\gamma}$	= mean void fraction
η	= efficiency
ρ	= density

Mathematical notation

A, B, C, D	= state space matrices
Z	= matrix
$f(\cdot)$	= continuous functions
$G(s)$	= transfer function
s	= Laplace transform variable
$\frac{d}{dt}$	= continuous derivative
Δ	= differential
u	= input signals
y	= output signals
x	= dynamic states
e	= external noise
a, b, c, k	= empirical parameters
τ	= time constant
T_d	= time delay
θ	= scheduling variable
$\forall$	= “for all”

Physical variables

A	= area
C_d	= discharge coefficient
L	= length
V	= volume
m	= mass
t	= time
λ	= normalized length
ω	= rotational velocity
C_p	= specific heat

Subscripts

$1, 2, \dots, j, \dots, n$	= first, second, etc. region
air	= air
amb	= ambient
$bulb$	= bulb
c	= condenser
cs	= cross-sectional
e	= evaporator
f	= liquid
g	= vapor
i	= in
int	= interface

<i>isen</i>	= isentropic
<i>k</i>	= compressor
<i>o</i>	= out
<i>r</i>	= refrigerant
<i>rec</i>	= receiver
<i>v</i>	= valve
<i>vol</i>	= volumetric
<i>w</i>	= wall

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